

2021 Test 2

Saturday, October 30, 2021 2:03 PM

Nanyang Technological University
SPMS/Division of Mathematical Sciences

2021/2022 Semester 1

MH2500 Probability & Introduction to Statistics

Test 2

Name &
Matriculation Number

Solutions

Score:

Tutorial Group:

9:00-10:20, 29/10/2021

1. (10 Points)

Let $X \sim \text{Exp}(\lambda)$.

(a) Find $\mathbb{E}(X^n)$, where $n \geq 1$.

(b) Find $\text{Var}(X)$.

(a) pdf of X : $\lambda e^{-\lambda x}$, $x \geq 0$

$$\mathbb{E}[X^n] = \int_0^{\infty} \underbrace{x^n}_{u} \underbrace{\lambda e^{-\lambda x}}_{v'} dx \quad \text{Apply Integration - By Parts -}$$

$$= \left[\frac{x^n \lambda e^{-\lambda x}}{-\lambda} \right]_0^{\infty} - \int_0^{\infty} \frac{n x^{n-1} \lambda e^{-\lambda x}}{-\lambda} dx$$

$$= 0 + \frac{n}{\lambda} \int_0^{\infty} \underbrace{x^{n-1}}_{u^{n-1}} \underbrace{\lambda e^{-\lambda x}}_{\text{pdf of } X} dx$$

$$= \frac{n}{\lambda} \mathbb{E}[X^{n-1}]$$

$$\therefore \mathbb{E}[X^n] = \frac{n}{\lambda} \times \frac{n-1}{\lambda} \times \frac{n-2}{\lambda} \times \dots \times \frac{2}{\lambda} \times \frac{1}{\lambda} = \frac{n!}{\lambda^n}$$

$$(b) \text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{2!}{\lambda^2} - \left(\frac{1!}{\lambda}\right)^2 = \frac{1}{\lambda^2}$$

2. (15 Points)

Suppose that of all the clouds that are seeded with silver iodide, 58% show splendid growth. If 60 clouds are seeded with silver iodide, find the probability that exactly 35 show splendid growth, by using normal distribution to approximate it.

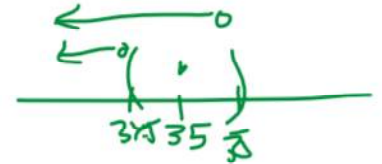
Let X be the r.v. denoting the number of clouds that show splendid growth.

$$X \sim \text{Bin}(60, 0.58) \quad E(X) = 60(0.58) = 34.8$$

$$\text{Var}(X) = 60(0.58)(1-0.58) = 14.616$$

Normal Approximation:

$$P(X=35) = P(34.5 < X < 35.5)$$



$$= P\left(\frac{34.5 - 34.8}{\sqrt{14.616}} < Z < \frac{35.5 - 34.8}{\sqrt{14.616}}\right)$$

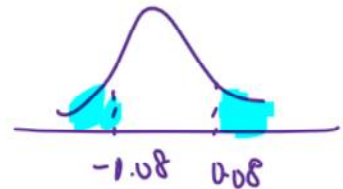
$$= P(-0.08 < Z < 0.18)$$

$$= \Phi(0.18) - \Phi(-0.08)$$

$$= 0.5714 - (1 - \Phi(0.08))$$

$$= 0.5714 - (1 - 0.5319)$$

$$\approx 0.1033$$

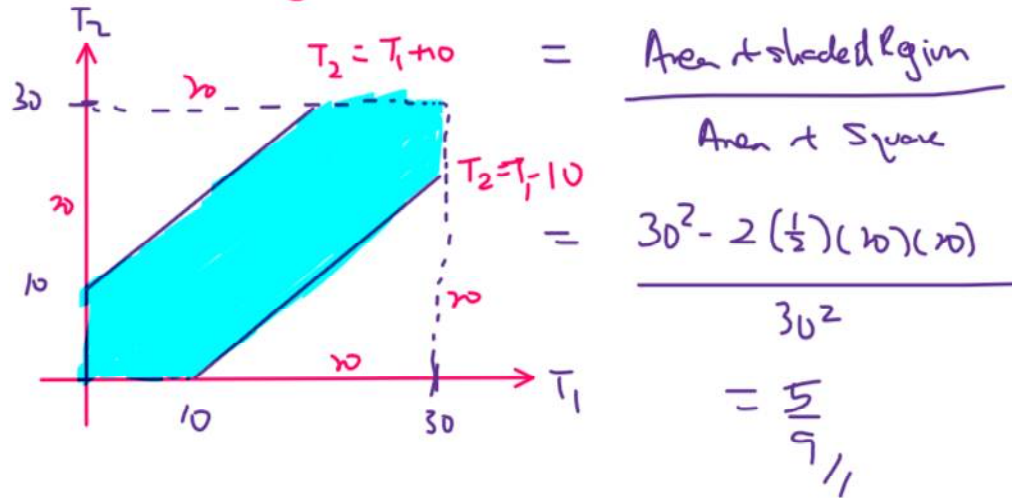


3. (15 Points)

Jack invites his friend to Canteen B for a coffee break, and they agreed to meet at SPMS lobby between 3:30pm to 4:00pm. If they arrive at random times during this period, what is the probability that they will meet within 10 minutes?

Let T_1 and T_2 be the times that Jack and his friend arrives at the lobby. $T_1 \sim \text{Unif}[0, 30]$, $T_2 \sim \text{Unif}[0, 30]$

$$P\{\text{They meet within 10 mins}\} = P(|T_1 - T_2| \leq 10)$$

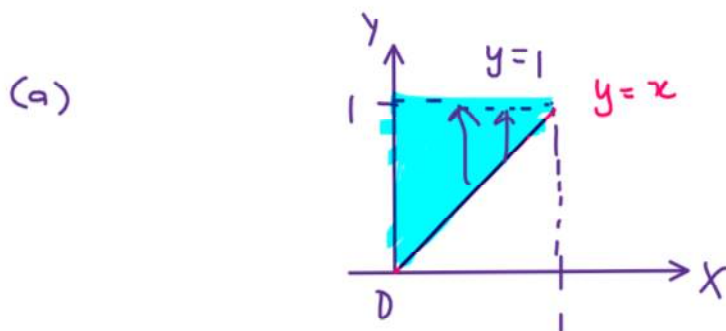


4. (20 Points)

The joint probability density function of random variables X and Y is given by

$$f(x, y) = \begin{cases} cxy^2 & \text{if } 0 < x < y < 1; \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Determine the value of c .
 (b) Find the marginal probability density function of X and Y .
 (c) Calculate $\mathbb{E}(X)$ and $\mathbb{E}(Y)$.



$$\int_0^1 \int_x^1 cxy^2 dy dx = 1$$

$$c \int_0^1 x \left[\frac{y^3}{3} \right]_x^1 dx = 1$$

$$\frac{c}{3} \int_0^1 x - x^4 dx = 1$$

$$\frac{c}{3} \left[\frac{x^2}{2} - \frac{x^5}{5} \right]_0^1 = 1$$

$$c = 10 //$$

$$\begin{aligned} \text{(b)} \quad f_X(x) &= \int_x^1 10xy^2 dy \\ &= 10x \left[\frac{y^3}{3} \right]_x^1 \end{aligned}$$

$$= \frac{10}{3} x (1 - x^3) \text{ if } 0 < x < 1 //$$

$$f_Y(y) = \int_0^y 10xy^2 dx = 10y^2 \int_0^y x dx = 10y^2 \left[\frac{x^2}{2} \right]_0^y$$

$$= 5y^4 \text{ } 0 < y < 1 //$$

$$\text{(c)} \quad \mathbb{E}(X) = \int_0^1 x f_X(x) dx \quad \mathbb{E}(Y) = \int_0^1 y f_Y(y) dy$$

$$= \int_0^1 \frac{10}{3} x^2 (1 - x^3) dx$$

$$= \int_0^1 5y^5 dy$$

$$= \frac{10}{3} \left[\frac{x^3}{3} - \frac{x^6}{6} \right]_0^1$$

$$= \left[\frac{5y^6}{6} \right]_0^1$$

$$= \frac{5}{9} //$$

$$= \frac{5}{6} //$$

5. (20 Points)

Let the conditional probability density function of X , given that $Y = y$, be

$$f_{X|Y}(x|y) = \frac{x+y}{1+y} e^{-x}, \quad 0 < x < \infty, \quad 0 < y < \infty.$$

(a) Find $P\{X < 1 | Y = 2\}$.

(b) Find $\mathbb{E}(X|Y = 2)$.

$$\begin{aligned}
 (a) \quad P\{X < 1 | Y = 2\} &= \int_0^1 \frac{x+2}{1+2} e^{-x} dx \\
 &= \frac{1}{3} \int_0^1 \underbrace{x}_{u} \underbrace{e^{-x}}_{v'} + 2e^{-x} dx \\
 &= \frac{1}{3} \left[\underbrace{[-xe^{-x}]}_0^1 - \int_0^1 -e^{-x} dx + \left[2e^{-x} \right]_0^1 \right] \\
 &= \frac{1}{3} \left[-e^{-1} + \left[-e^{-x} \right]_0^1 + \left[-2e^{-x} \right]_0^1 \right] \\
 &= \frac{1}{3} \left[-e^{-1} - 3(e^{-1} - 1) \right] \\
 &= \frac{1}{3} (3 - 4e^{-1}) //
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad \mathbb{E}[X|Y=2] &= \int_0^{\infty} x \left(\frac{x+2}{1+2} e^{-x} \right) dx \\
 &= \frac{1}{3} \left[\int_0^{\infty} \underbrace{x^2}_{u} \underbrace{e^{-x}}_{v'} dx + \int_0^{\infty} \underbrace{2x}_{u} \underbrace{e^{-x}}_{v'} dx \right] \\
 &\quad \vdots \leftarrow \text{Exercise...} \\
 &= \frac{1}{3} \left[2! + 2(1!) \right] \\
 &= \frac{4}{3} //
 \end{aligned}$$

6. (20 Points)

Let $X_1, X_2, \dots, X_n$ be independent random variables with a common cdf F and density f , and let U, V denote the *maximum* and the *minimum* of the X_i 's, respectively. That is, $U = \max\{X_1, X_2, \dots, X_n\}$ and $V = \min\{X_1, X_2, \dots, X_n\}$.

- (a) Find the cumulative distributive function F_U and the probability density function f_U for U .
- (b) Find the cumulative distributive function F_V and the probability density function f_V for V .
- (c) Suppose that $Y_1, Y_2, \dots, Y_n \sim \text{Exp}(\lambda)$ are independent. Find the probability density function for the variable $\min\{Y_1, Y_2, \dots, Y_n\}$.

$$\begin{aligned}
 (a) \quad F_U(u) &= P(U \leq u) \\
 &= P(X_1 \leq u, X_2 \leq u, \dots, X_n \leq u) \\
 &\stackrel{\text{independence}}{=} P\{X_1 \leq u\} P\{X_2 \leq u\} \cdots P\{X_n \leq u\} \\
 &= [F(u)]^n, \\
 f_U(u) &= \frac{d}{du} [F(u)]^n = n [F(u)]^{n-1} f(u)
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad F_V(v) &= P(V \leq v) \\
 &= 1 - P(V > v) \\
 &= 1 - P(X_1 > v) P(X_2 > v) \cdots P(X_n > v) \\
 &= 1 - (1 - F(v))^n,
 \end{aligned}$$

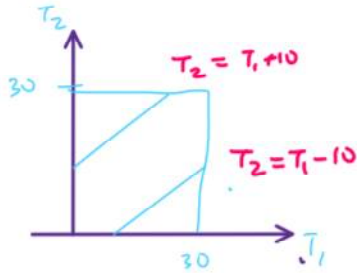
$$\begin{aligned}
 f_V(v) &= \frac{d}{dv} \{1 - (1 - F(v))^n\} = -n (1 - F(v))^{n-1} \{-f(v)\} \\
 &= n (1 - F(v))^{n-1} f(v),
 \end{aligned}$$

$$\begin{aligned}
 (c) \quad Y_i &\sim \text{Exp}(\lambda), \quad f_Y(y) = \lambda e^{-\lambda y} \text{ if } y \geq 0 \\
 F(y) &= 1 - e^{-\lambda y}
 \end{aligned}$$

$$F_Y(y) = 1 - e^{-\lambda y}$$

$$\begin{aligned} F_{\min}(D), f_V(v) &= n [1 - (1 - e^{-\lambda y})]^{n-1} \lambda e^{-\lambda y} \\ &= n \lambda e^{-\lambda y(n-1)} e^{-\lambda y} \\ &= n \lambda e^{-n\lambda y}, \text{ if } y \geq 0. \quad \therefore V \sim \text{Exp}(n\lambda). \end{aligned}$$

EXTRA: Calculus(!) Approach for Q3. (Unnecessary, but u can try this approach if u like)



Consider $0 \leq T_2 \leq 10$, $10 \leq T_2 \leq 20$, $T_2 \geq 20$.

$$\begin{aligned} &\int_0^{10} \int_0^{T_2+10} \frac{1}{900} dT_1 dT_2 \\ &= \frac{1}{900} \int_0^{10} [T_1]_0^{T_2+10} dT_2 \\ &= \frac{1}{900} \int_0^{10} T_2 + 10 dT_2 \\ &= \frac{1}{900} \left[\frac{1}{2} T_2^2 + 10 T_2 \right]_0^{10} \\ &= \frac{1}{900} \left[\frac{1}{2} (10)^2 + 100 \right] \\ &= \frac{150}{900} \end{aligned}$$

$$\int_{10}^{20} \int_{T_2-10}^{T_2+10} \frac{1}{900} dT_1 dT_2$$

$$\begin{aligned} &= \frac{1}{900} \int_{10}^{20} 20 dT_2 \\ &= \frac{1}{900} (20)(10) \\ &= \frac{200}{900} \end{aligned}$$

$$\int_{20}^{30} \int_{T_2-10}^{30} \frac{1}{900} dT_1 dT_2$$

$$\begin{aligned} &= \frac{1}{900} \int_{20}^{30} 40 - T_2 dT_2 \\ &= \frac{1}{900} \left[40 T_2 - \frac{1}{2} T_2^2 \right]_{20}^{30} \\ &= \frac{1}{900} [(1200 - 450) - (800 - 200)] \\ &= \frac{150}{900} \end{aligned}$$

$$\frac{150}{900} + \frac{200}{900} + \frac{150}{900} = \frac{5}{9}$$

Complement:

$$\begin{aligned} &1 - \int_0^{20} \int_{T_1+10}^{30} \frac{1}{900} dT_2 dT_1 - \int_{10}^{30} \int_0^{T_1-10} \frac{1}{900} dT_2 dT_1 \\ &= 1 - \frac{1}{900} \int_0^{20} 20 - T_1 dT_1 - \frac{1}{900} \int_{10}^{30} T_1 - 10 dT_1 \\ &= 1 - \frac{1}{900} \left[20 T_1 - \frac{1}{2} T_1^2 \right]_0^{20} - \frac{1}{900} \left[\frac{T_1^2}{2} - 10 T_1 \right]_{10}^{30} \\ &= 1 - \frac{1}{900} (200) - \frac{1}{900} ((450 - 300) - (50 - 100)) \\ &= 1 - \frac{2}{9} - \frac{2}{9} = \frac{5}{9}. \end{aligned}$$